

SOFTWARE DEVELOPMENT

Extensive experience in executing full life-cycle development projects; ramping-up projects within time, budget & quality parameters, as per project management & best practice guidelines

Objective:

Pursuing a lead position in Embedded industry with 7+years of experience in automotive domain with classic autosar (Borg-Warner), I can utilise and build upon my current knowledge of modern Embedded Innovation, design and engineering. There by approaching a positive growth for my career along with the cherished reputation of the organisation

PROFESSIONAL SKILL

PROFESSIONAL EXPERIENCE (BorgWarner PDS) (SEP 2020 – August 2025) (4 YR ,11 MONTH)

- ◆ Working Experience in IDM -TAM INVERTER ECU for HMC-KIA(Classic Autosar).
- ◆ Direct interaction with Customers for requirement analysis based on project requirements.
- ◆ Strong Knowledge in Automotive **Communication Protocol CAN and SPI, I2C**
- ◆ Working Experience in **Embedded C / C++ , Assembly language** development .
- ◆ Working Experience in **COM Stack, Diagnostic Stack.**
- ◆ Working Experience in **RTE and BSW Development.**
- ◆ Working Experience in **OS Modules configuration, design,Core-partitioning.**
- ◆ Working Experience in **Mode Management(sleep, startup, shutdown strategy).**
- ◆ Working Experience in fixing issue for **RTE layer**, Application layer along with **BSW configuration** in both **Da Vinci Developer** and configurator.
- ◆ Having Experience in Development of **RTE Interfaces with COM Modules, NVM Modules, Diagnostic Modules.**
- ◆ Working Experience with **BSW Modules like COM, NVM and DCM & DEM OS configurations, Code changes issue fixes.**
- ◆ **Working Experience in implementing Interface in RTE (Sender Receiver Interface, Client-Server Interface, Mode-Switch Interface, and NV Data Interface)**
- ◆ Working Experience in **signal Mapping from Da Vinci (Developer, Configurator) Tool for Autosar Configuration in RTE and BSW.**
- ◆ Good Experience in **Make File** Development for Integration of all modules.
- ◆ Good Understanding of **Microcontroller Architecture** design of **Infineon/Aurix TC 37XXX(Quad core-controller).**
- ◆ Having experience in Software Development Life Cycle in **AGILE Methodologies.**
- ◆ Working Experience on **DCM ,DEM MODULES** implementation of **UDS and OBD** services **DIDs & RIDs.**
- ◆ Debugging software issues reported from customer/System team and provide timely fixes.
- ◆ Experience in Debugging Tool like **TASKING Compiler and Lauterbach, BitBucket, Trace32 Canoe ,CAN Analyser.**
- ◆ Analyse and review tickets from JIRA and test them on static/real load benches.
- ◆ Familiar with **AUTOSAR standards** and working experience with **R3.1.4 and 4.2.2**
- ◆ Good knowledge in **QNX RTOS Concept.**
- ◆ Working Experience in **GitHub Copilot AI and CodeGeek** using **ai generative code generation.**

PROFESSIONAL EXPERIENCE

- ◆ Having Working Experience in Vehicle Simulation ,Tool **CANoe** and **CAN Analyser,Canapé.**
- ◆ Experience in **RTE Generation, BSW Development and Autosar Configuration.**
- ◆ Experience with **ASPICE** compliant software development worked for **SWE.1 to SWE.6.**
- ◆ Working Experience in **Integration of third-Party Component Integration. (BSP Integration in Autosar Stack).**
- ◆ working Experience with global teams (Singapore,Germany and US) on software projects.
- ◆ Working in Tracking Tools like **Confluence Atlassian/ jira, version control tools, like git, Plastic(SCM).**
- ◆ Having Experience and working in vehicle simulation tool(**HIL bench**), **CAN Analyser.**

- ✦ Experience in Application Development in AUTOSAR.
- ✦ Good knowledge in Debugging code and fixing the defects of Application code.
- ✦ Good Experience with customer interaction/handling.
- ✦ Good Experience in process management tool Polarion, Share-point work product maintaining.

REAL TIME EMBEDDED SKILL

- ✦ Analysing & Software Requirements/Specifications (SRS)
- ✦ Data flow diagrams, Designing & Developing.
- ✦ Designing & Developing
- ✦ Communication Protocol.
- ✦ Plans/Procedure. Unit Testing. Communication Protocol.
- ✦ Software Design Documentation (SDD). Software Development Test Plans (SDTP) Software Design Test Result (SDTR)

TECHNICAL SKILLS

- ✦ Programming Languages. : Embedded C ,C++ , Assembly Language.
- ✦ Server : Bit bucket, plastic, Git, BitBucket .
- ✦ Processor Architecture. : Aurix Tricore Tc37xxx
- ✦ Protocols : CAN,I2C, SPI,AUTOSAR
- ✦ Development Tool : Davinci (Developer,Configurator),Canoe, Polyspace, Lauterbach, Trace32

EDUCATIONAL BACKGROUND

- ✦ Graduated as B.Tech in stream of EEE (Electrical and Electronic Engg) from Gandhi Institute Technological Advancement College, Bhubaneswar under B.P.U.T. University ROURKELA.
- ✦ Completed Intermediate with Science from the B.D public (C.B.S.E)
- ✦ Completed S.S.C from the Shivam High School, PATNA (H.S.E) of Bihar.

PROJECTS

PROJECT # 1 : HMC(IDM) INVERTER (BorgWarner PDS) (*SEP 2020 – 21st AUGUST 2025*)(4YR ,11 MONTH)
Power Drive train project : **AUDI-TSE GEN, PEGASUS, STELLANTIS (DTCM AND DT & DJ PROJECT)**
, KMDTCM , M182(Maserati)

ROLE : Senior Software Developer

ENVIRONMENT : Windows 10,Notepad++, Beyond Compare, Infineon Evaluation Board (Aurix tc37xx controller), CAN Analyser, Da Vinci (developer & configurator) , Canoe, Polyspace, **Lauterbach Trace32**

Roles And Responsibilities:

- ✦ Data flow diagrams, Designing & Developing.
- ✦ Design and implementing the whole feature for the different variant.
- ✦ Coding, Unit Testing, Debugging.
- ✦ Integrated software for RTE, BSW.
- ✦ Creation of application SWC and mapping runnable, port and interface and Resolving issue based on errors at the initial levels for developer and configurator.
- ✦ Import multiple DBC files and
- ✦ Handling the Integration Issue.
- ✦ make file preparation.
- ✦ Handling the configuration of DEM Module and DCM Module.

- ✦ Implementation of UDS SERVICES Rx and Tx messages along with diagnostics 10, 11, 22, 2E, 31, 27, 28, 04, 14, 85.
- ✦ Implementation of DTCs , events, enable conditions, freeze-frame data, extended data, DTC NVM records, DTC suppression feature.
- ✦ Handling the DCM and DEM issues Independently writing test Report.
- ✦ Preparing Design Template.
- ✦ Implementation of OS task, Runnable.
- ✦ Implementation of OS SC3 memory protection and memory mapping.
- ✦ Implementation of RTE ports and connections.
- ✦ Software Design Documentation (SDD) over Polarian.
- ✦ Preparation of SFT (SOFTWARE FUNCTIONALITY TEST REPORT)
- ✦ Handling Integrated Test Plans (ITP) and Integrated Test Report (ITR).
- ✦ Software development according to ISO26262 and ASPICE standards.
- ✦ Improving Existing software.
- ✦ Develop documentation throughout the software development life cycle(SDLC).
- ✦ collaborate with internal teams to produce software design and architecture.
- ✦ Configuration and integration of Autosar BSW component like Nvm ,COM, Canif, PduR, OS, DCM, DEM, RTE ports, NVM ,BSWM etc.
- ✦ Analysing the QAC and Fixing SCA warning POLYSPACE TOOLS as per FCA rules sets for mandatory and Required warnings.

PROJECT # 4 : AUTOSAR 4.2.1 (Baxter R&D)(RANGAM INFO TECH)(AUG 2019 – DEC 2019)

ROLE : SOFTWARE ENGINEER

ENVIRONMENT : Windows 10,Notepad++, Beyond Compare, CAN Analyser,Davinci developer, Davinci configurator , Canoe, polyspace, Lauterbach Trace32 .

Implementation Modules in AUTOSAR 4.2.1:

- ✦ Implementation in BSW with NVM.
- ✦ Implementation in RTE with NVM.
- ✦ Interaction of BSW component via RTE to application

Roles and Responsibilities:

- ✦ Analyzing & Software Requirements/Specifications (SRS and SWS).
- ✦ Data flow diagrams, Designing & Developing.
- ✦ Design and implementing the whole feature for the different variant.
- ✦ Coding. Unit Testing, Debugging.
- ✦ Handling the Integration Issue.
- ✦ Handling independently the module.
- ✦ Preparing Design Template.
- ✦ Software Design Documentation (SDD). Software Development Test Plans (SDTP). Software Development Test Plans (SDTP).

PROJECT # 2 : COM DIAGNOSTIC (GE-Transportation) (MARCH 2018 – MAY 2019)

ROLE : SENIOR SOFTWARE ENGINEER

ENVIRONMENT : Windows 10,Notepad++,ECU Worx, Beyond Compare, Lab Car, CAN Analyser, Tasking Compiler, Da Vinci developer and Configurator, Canoe, Polyspace, Lauterbach Trace32 .

Roles And Responsibilities:

- ✦ Analyzing the Software Requirements/Specifications.
- ✦ Develop the Requirements providing from Costumer.
- ✦ Build and compile the software in tasking compiler.
- ✦ Flashing the integrating Software through tool
- ✦ Doing the Software Analysis according to the MISRA C guidelines
- ✦ Debugging the code and fixing the issues.
- ✦ Doing the Root cause Analysis for different types of defects.
- ✦ Preparing Test Report, Test Case, Test Log for Every Release

PROFESSIONAL EXPERIENCE

Open for Onsite jobs as well.

Organization : MARELLI (Suitable ai) (infotainment and cluster with Battery management).
(SEP 2025 - NOVEMBER 2025)

Designation : LEAD Software Engg(C2H EMPLOYEE)
Duration : 03 Months

Organization : BorgWarner (Power drive system).(SEP 2020 – 21ST AUGUST 2025)

Designation : Senior Software Developer (PERMANENT EMPLOYEE)
Duration : 4 year 11 Months (gap due to covid before joining Borgwarner)

Organization : BAXTER (RANGAM INFO TECH).(AUG 2019 – DEC 2019)

Designation : Software Developer (CONTRACT EMPLOYEE)
Duration : 6 Months

Organization : GENERAL ELECTRIC (QUEST GLOBAL INDIA LTD.)(MARCH 2018 – MAY 2019)

Designation : Software Developer (CONTRACT EMPLOYEE)
Duration : 1.2 Year

I hereby declare that the above information provided in this resume is true and accurate to the best of my knowledge. I also declare that I am having the necessary documents to prove my eligibility.